ASEAN
GEOAI FUSION 2026

Module 7: Advanced Knowledge: Advanced Geospatial Techniques in GeoAI Applications (AD1002)

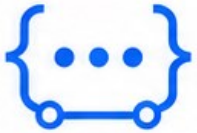
AGAIF 2026 Boot Camp 2
Level: Advanced Knowledge

23 July 2026



Session 4:
Model Deployment Strategies for GeoAI and Deep Learning Applications

Session Agenda



1. API Development



2. Model Serving Platforms



3. Monitoring Model Performance



4. Deployment of Deep Learning Models for GeoAI Applications



5. Example: Deploying CNN-Based Satellite Image Classification or LSTM-Based Network Forecasting Models

Learning Outcomes

1. Explain why a trained model must be deployed.
2. Describe how an API connects a model to other applications.
3. Compare common model-serving and deployment options.
4. Identify the main elements used to monitor a deployed model.
5. Explain how the LSTM-RSRP model can be deployed as a GeoAI service.

API Development

An API is a communication link between an AI model and another application.

Application → API → AI model → Prediction → Application

1 APPLICATION



Mobile app sends new network data

2 API RECEIVES REQUEST



API validates and carries the input

3 AI MODEL



Deployed LSTM processes the observations

4 PREDICTION



Model produces the next RSRP

5 API RETURNS RESULT



Application displays the prediction

Think of the API as a waiter:



Application = customer places an order



API = waiter carries the order



AI model = kitchen prepares the result



API = waiter returns the result

LSTM-RSRP EXAMPLE

Previous 10 network observations



RSRP • signal quality • SNR • CQI • bitrate • speed • network type • location

POST /predict-rsrp

Endpoint that receives the previous 10 observations and returns the predicted next RSRP.

Predicted next RSRP

-78 dBm

GET /health

Is the API running?

GET /version

Which model produced the result?

The API does not make the prediction. It carries the input, calls the model and returns the model output.

Model-Serving and Deployment Options

Model serving keeps a trained model ready to receive data and return predictions.

Application sends data → Model processes data → Prediction is returned

Option	Simple explanation	Best used for	Example
FastAPI	Creates an API for a Python model	Learning and small applications	RSRP prediction endpoint
TensorFlow Serving	Runs saved TensorFlow models	TensorFlow production systems	Serve many prediction requests
MLflow Serving	Serves models and manages versions	Testing and model tracking	Use model version 2
Docker	Packages the model, code and libraries	Running the same setup anywhere	Move from laptop to cloud
Managed Cloud	Hosts, scales and monitors the model	Applications with many users	Public GeoAI dashboard
Edge Device	Runs the model near the data source	Fast or offline prediction	Prediction at a cell tower

Simple LSTM-RSRP Example

For a student project, use FastAPI to create a prediction endpoint. Package the API and model with Docker, then run them on a small cloud server.

Dashboard sends 10 network observations → FastAPI calls the LSTM → Dashboard receives the predicted next RSRP

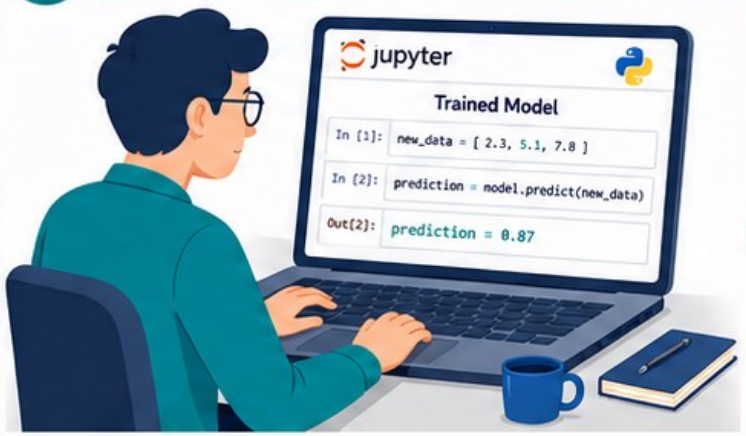
FastAPI, TensorFlow Serving and MLflow provide model access. Docker packages the system. Cloud and edge describe where it runs.

Choose based on model framework, number of users, required speed, internet connectivity and available resources.

What Is Model Deployment?

Model deployment means making a trained model available for real applications to use.

1 BEFORE: MODEL IN A NOTEBOOK



Only the researcher can use it by running Python code.

Enter data manually → Run the model → See tthe prediction in Python

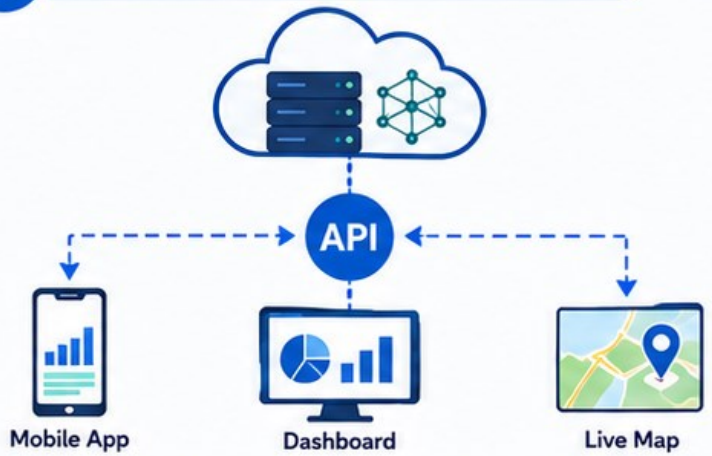
2 DEPLOY THE MODEL



The trained model is saved, hosted and connected to an API.

API = a connection that lets another application send data to the model and receive a prediction.

3 AFTER: MODEL AS A SERVICE



Applications can use the model without opening the notebook.

Application sends new data → Model makes a prediction → Application displays the result

Example: A network dashboard sends recent RSRP values. The deployed LSTM returns the predicted next RSRP.

Training builds the model. Deployment makes the model usable in a real system.

Deployment changes how the model is accessed; it does not automatically improve accuracy.

Monitoring Model Performance

Model monitoring continuously checks whether a deployed model and its supporting service are working correctly.

1 SYSTEM

Is the service working?

Check: Response time • requests • errors • availability • memory

Warning: the API response becomes too slow.

Deployed LSTM-RSRP Service



Predicts next RSRP

Logs & metrics

Alerts & insights

2 INPUT DATA

Is the incoming data complete and valid?

Check: Missing values • data types • ranges • sequence length

Warning: only 4 observations are sent instead of 10.

3 MODEL PERFORMANCE

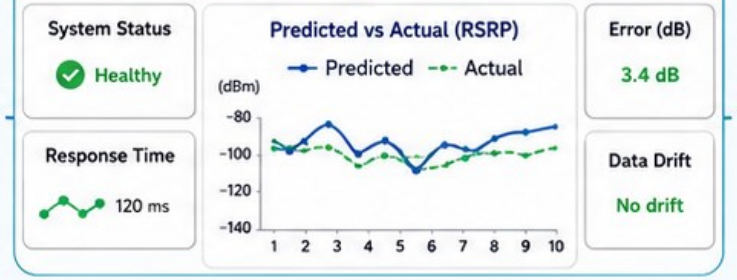
Are the predictions still accurate?

Check: MAE • RMSE • error by route, operator or technology



Predicted: −99.6 dBm
Actual: −103.0 dBm
Error: 3.4 dB

Monitoring Dashboard



4 DATA DRIFT

Has real network data changed?

Check: Compare new data with the training data

Warning: a new route or frequency band appears.

Real network conditions change. A model that worked well during training may become less reliable after deployment.

What to do when performance declines

1. Check the new input and preprocessing
2. Compare new data with the training data
3. Retrain and test the model if needed
4. Deploy the validated update or return to a stable version

A deployed model must be monitored because real network conditions can change.

Choosing a GeoAI Deployment Strategy

Choose based on how quickly the result is needed and where the model should run.

1 BATCH

Can the result wait?

Processes many records together at scheduled times.



GeoAI example: Generate a network coverage map every night.

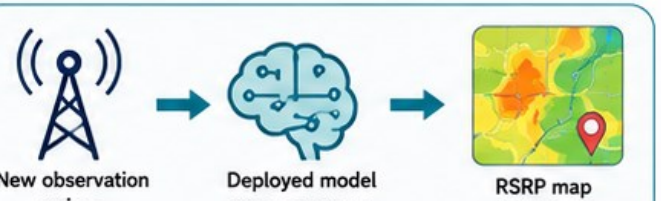
Best when: Immediate action is unnecessary.

Main advantage: Efficient for historical analysis and planning.

2 REAL TIME

Is a decision needed now?

Predicts immediately when new data arrives.



GeoAI example: Update the RSRP map as new observations arrive.

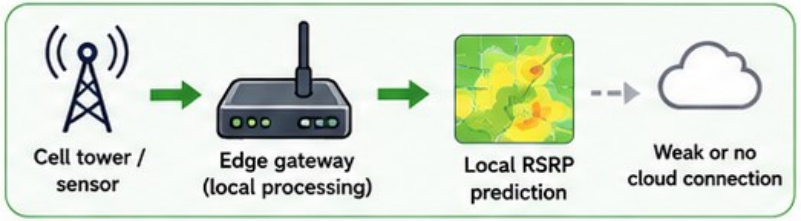
Best when: The prediction supports a current decision.

Main advantage: Fast response to changing conditions.

3 EDGE

Must processing happen near the data source?

Runs the model locally on a gateway or network device.



GeoAI example: Predict RSRP locally at a cell-tower gateway.

Best when: Low latency, weak connectivity or limited data transfer is important.

Main advantage: Fast local results with less cloud communication.

DECISION GUIDE

Can wait? → BATCH

Need the result now? → REAL TIME

Need fast local processing or limited internet? → EDGE

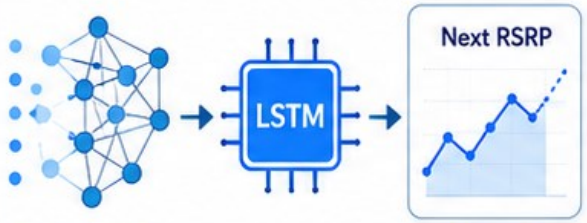
The same trained model may use different deployment strategies for different operational needs.

Batch processes later. **Real time** responds now. **Edge** processes nearby.

From Prediction to Real-World Deployment

SESSION 2 — PREDICT

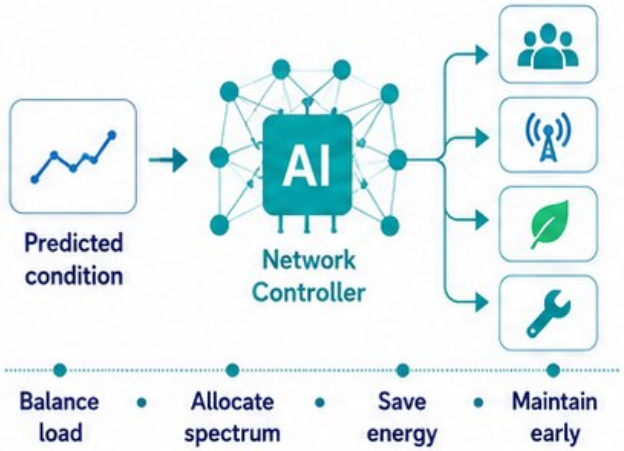
What may happen next?



The LSTM predicted the next RSRP from previous network observations.

SESSION 3 — DECIDE

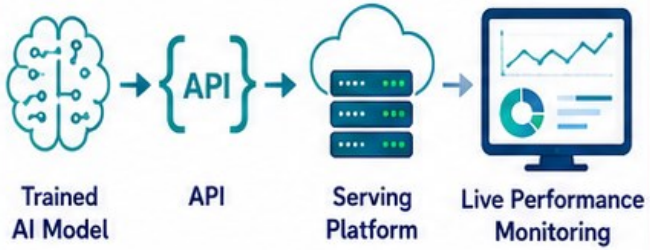
What should the network do?



AI used the predicted condition to support network optimisation decisions.

SESSION 4 — DEPLOY

How can the model operate in a real system?



Expose the model through an API, host it in a serving environment and monitor it continuously.

Predict what may happen → **Decide what to do** → **Deploy and monitor the model**

LSTM–RSRP GeoAI Deployment Example

Use the latest 10 network observations to predict the next RSRP and show it on a map.

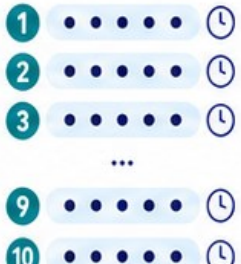
1 COLLECT



Receive network measurements: RSRP, signal quality, speed and location.

2 KEEP 10

Store the latest 10 observations in time order.



3 SEND TO API

The application sends the sequence to POST /predict-rsrp.



4 PREPARE DATA

Use the same cleaning, scaling and input shape used during training.



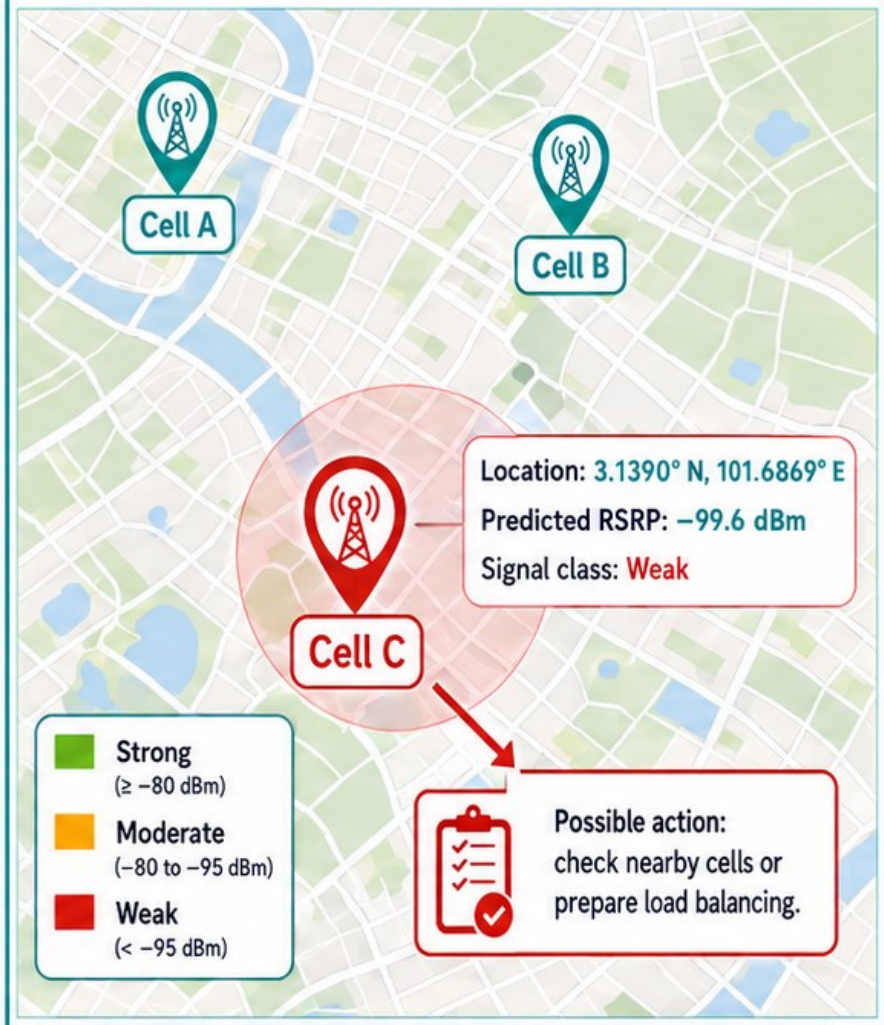
5 LSTM PREDICTS

The model estimates the next RSRP.



Predicted RSRP: −99.6 dBm

6 SHOW THE PREDICTION ON A GEOAI MAP



GeoAI links the prediction to WHERE it occurs.

1 MODEL PREDICTS

The next RSRP will be

−99.6 dBm

2 ACTUAL VALUE ARRIVES

The network later records

−103.0 dBm

3 CALCULATE THE ERROR

Use absolute error (in dB):

$|-99.6 - (-103.0)| = 3.4$ dB

4 WHAT DOES IT MEAN?

The model predicted a signal **3.4 dB stronger** than the value recorded.

RSRP values are negative. A value closer to zero is stronger, so −99.6 dBm is stronger than −103.0 dBm.

Store this error and repeat the comparison for many predictions. One result alone is not enough to judge the model.

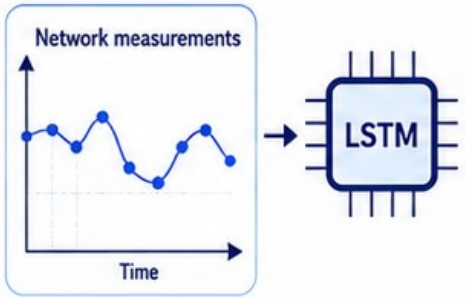
Time-series model predicts WHAT and WHEN → GeoAI shows WHERE → the network decides WHAT TO DO

Summary: How Sessions 2–4 Connect

The same prediction moves from analysis to action and operation.

1 SESSION 2 — PREDICT

What may happen next?

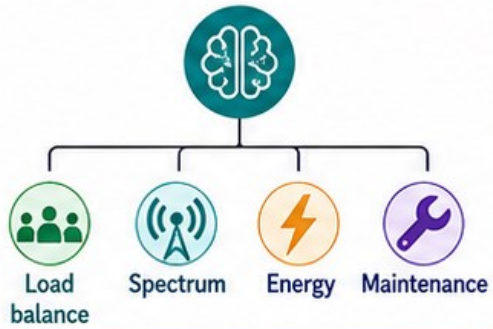


LSTM predicts that Cell C may have weak RSRP.

Prediction

2 SESSION 3 — DECIDE

What should the network do?

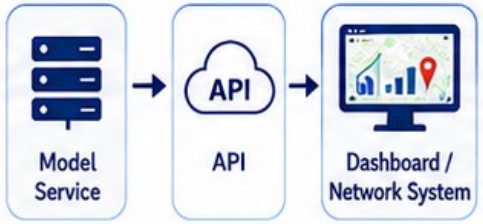


AI may balance traffic, adjust spectrum, save energy or plan maintenance.

Recommended action

3 SESSION 4 — DEPLOY

How is the model used in a real system?

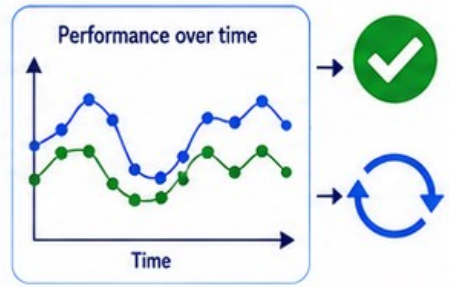


An API sends the prediction to a dashboard or network system.

Delivered to system

4 MONITOR — KEEP IT RELIABLE

Is the deployed model still working well?



Compare predictions with actual values, check errors and update the model when needed.

GeoAI adds location:
it shows **WHERE** the predicted network condition occurs.



Session 2 predicts → Session 3 decides → Session 4 deploys and monitors

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THANK YOU

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